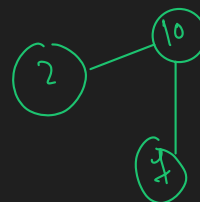
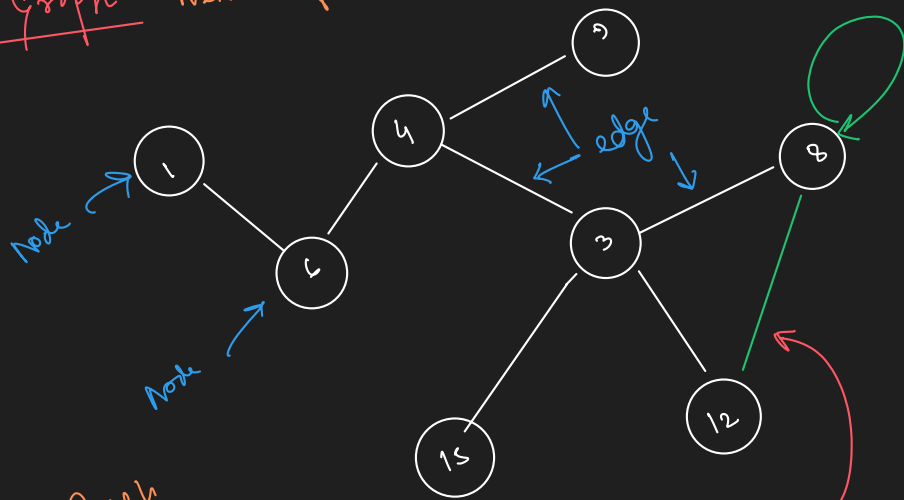


Day-1st

Graph Network of Nodes



Unweighted Graph

Directed Graph

Uni-directed



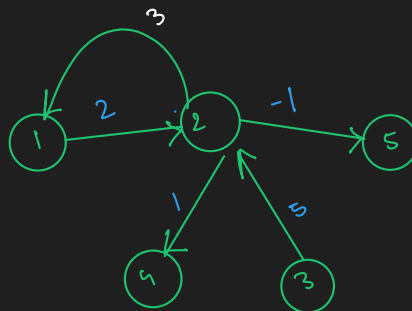
bi-directed



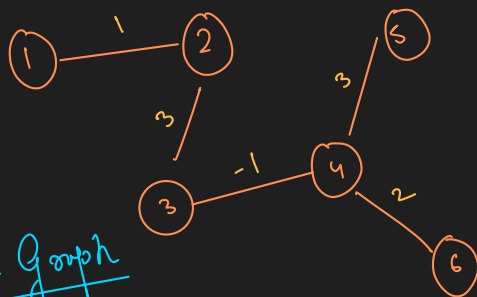
Sum



Undirected Graph



Weighted Graph



Storing a Graph

✓ Adjacency List

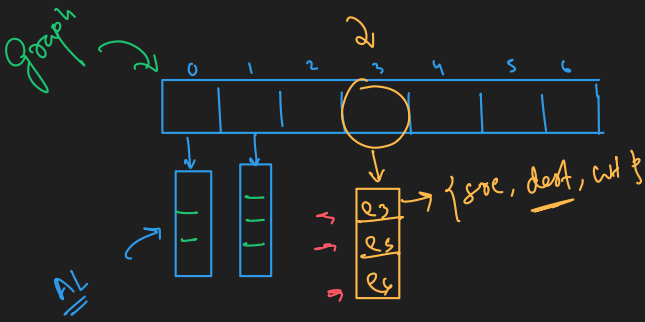
• Adjacency Matrix ✓

• Edge List ← Input

• 2D MxN [implicit graph] ← part of Question

1	0	1	0
0	0	1	0
0	1	1	1
0	0	0	1

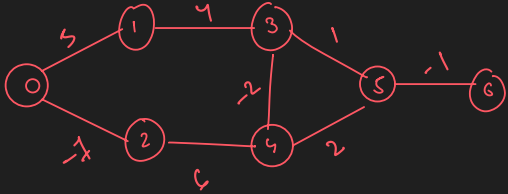




dest type

```
ArrayList<Edge> graph[] = new ArrayList[V]
```

```
for(int i=0 to graph.length){
    graph[i] = new ArrayList<>();
}
```



```
static class Edge{
    int src;
    int dest;
    int wt;

    public Edge(int src, int dest, int wt){
        this.src = src;
        this.dest = dest;
        this.wt = wt;
    }
}
```

Find Nearest Neighbour . (3)

```
public static void nearestNeighbour(ArrayList<Edge>[] graph, int ver){
    for(int i=0; i<graph[ver].size(); i++){
        Edge curr = graph[ver].get(i);
        System.out.println(curr.dest);
    }
}
```

```
public static void main(String[] args) {
    int V = 7;
    @SuppressWarnings("unchecked")
    ArrayList<Edge> graph[] = new ArrayList[V];

    createGraph(graph);

    // Nearest Neighbour
    int ver = 6;
    nearestNeighbour(graph, ver);
}
```

```
public static void createGraph(ArrayList<Edge>[] graph){
    for(int i=0; i<graph.length; i++){
        graph[i] = new ArrayList<>();

        graph[0].add(new Edge(src: 1, dest: 0, wt: 5));
        graph[0].add(new Edge(src: 0, dest: 2, wt: -1));

        graph[1].add(new Edge(src: 0, dest: 1, wt: 5));
        graph[1].add(new Edge(src: 1, dest: 3, wt: 4));

        graph[2].add(new Edge(src: 2, dest: 0, wt: -1));
        graph[2].add(new Edge(src: 2, dest: 4, wt: 6));

        graph[3].add(new Edge(src: 3, dest: 1, wt: 4));
        graph[3].add(new Edge(src: 3, dest: 4, wt: -2));
        graph[3].add(new Edge(src: 3, dest: 5, wt: 1));

        graph[4].add(new Edge(src: 4, dest: 2, wt: 6));
        graph[4].add(new Edge(src: 4, dest: 3, wt: -2));
        graph[4].add(new Edge(src: 4, dest: 5, wt: 2));

        graph[5].add(new Edge(src: 5, dest: 3, wt: 1));
        graph[5].add(new Edge(src: 5, dest: 4, wt: 2));
        graph[5].add(new Edge(src: 5, dest: 6, wt: -1));

        graph[6].add(new Edge(src: 6, dest: 5, wt: -1));
    }
}
```